

NSP2-S1-25-EC3_SP-SPMAIMLCZG515_DISTRIBUTED MACHINE LEARNING

RAVI SHANKAR SAHAY | 21 Jun 2026



Question-Wise Details

Section 1
Section #1

6
question(s)

1h 53m 13s
Time taken

Q
1

▼ Question 1

🕒 Time taken: 21m 27s

Question

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a) A healthcare technology company is developing an AI-based diagnostic system to classify patient records as **Disease: Positive** or **Disease: Negative**, with the goal of achieving higher predictive accuracy. Three independent teams evaluated different machine learning approaches on the same dataset, yielding the following results:

Team	ML Model proposed	Accuracy
A	Naïve Bayes classifier	60%
B	Random Forest classifier	62%
C	SVM classifier	58%

Based on the above scenario, answer the following:

- i) Analyze and explain the potential factors contributing to the modest accuracy levels observed across all three models. Consider aspects such as data quality, class imbalance, feature engineering, and model limitations. **[1 Mark]**
- ii) Critically evaluate whether it is feasible to develop a superior AI model that outperforms all three proposed solutions. Support your answer with appropriate justification. **[1 Mark]**
- iii) If improvement is achievable, recommend a more effective AI model or architecture, explaining why it would likely yield better diagnostic performance in this medical classification context. **[1 Marks]**

b) A deep learning engineer has proposed the following Architecture for a multi class classification task. Carefully examine the architecture specification given below and validate its design choices, identifying any flaws, inconsistencies, or improvements needed. **[2 Marks]**

Layer	Nodes	Activation function
Input Layer	8	-
Hidden Layer 1	5	Tanh
Hidden Layer 2	10	Sigmoid
Hidden Layer 3	3	Tanh
Output Layer	4	Softmax

Response

Words : 0

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Q.
2

▼ Question 2

🕒 Time taken: 12m 35s

Question

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A binary classification dataset is presented below, where X_1 and X_2 are input features and Y is the target output label: [5 Marks]

S. No	X_1	X_2	Y (Target Variable)
1	1	0	1
2	1	1	0
3	0	1	1
4	0	0	0

- Propose a suitable Perceptron architecture to correctly classify this dataset.
- Justify why a Single Layer Perceptron (SLP) would or would not suffice.
- If an MLP is required, draw and describe the complete architecture with activation functions.
- For your proposed architecture, calculate the total number of trainable parameters.

Response

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Question

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Consider the following Python code snippet and answer the questions followed.

[5 Marks]

```
model = Sequential()
model.add(Dense(64, input_dim=30, activation='relu'))
model.add(Dense(32, activation='relu'))
model.add(Dense(1, activation='sigmoid'))

model.compile(loss='binary_crossentropy', optimizer='SGD', metrics=['accuracy'])
model.fit(X_train, y_train, epochs=20, batch_size=16)

test_loss, test_acc = model.evaluate(X_test, y_test)
print(f"Test Loss: {test_loss}, Test Accuracy: {test_acc}")
```

- a) Draw the architecture related to the above code snippet.
- b) Nature of the problem being solved (Binary / Multi-class Classification or Regression).
- c) Why is the Sigmoid activation function used in the output layer instead of ReLU or Softmax? Under what condition would you replace Sigmoid with Softmax at the output layer?
- d) Calculate the number of trainable parameters (weights + biases) for each layer in the network.
- e) Why is binary cross-entropy chosen as the loss function?
- f) What does an epoch represent in the training process?
- g) Suppose the model outputs a prediction value of 0.73 for a particular instance. Then discuss the inference of the model output.

Response

Words : 0

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Question

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A Deep Neural Network is being trained to predict house prices based on three features: [5 Marks]

X1 — Area of the House (sq. ft.),

X2 — Number of Bedrooms

X3 — Distance from City Center (km)

After one forward pass and backpropagation on a batch of training samples, the computed gradient is: $\nabla J = [10, 40, 8, 20]$,

Input vector X(after normalisation) = [1, 0.6, 0.2, 0.3]

The current weights (including bias w_0) are: $W = [w_0, w_1, w_2, w_3] = [10, 5, 3, 7]$

i) Perform one gradient descent weight update using $\alpha=0.001$. Show the updated value of all four weights with complete calculations.

ii) Repeat the weight update from the same initial weights using $\alpha=0.9$. Show all four updated weight values.

iii) Which learning rate produces a larger weight update?

iv) What potential problems arise when using a very large learning rate?

v) Discuss all possible convergence criteria of Gradient descent computations.

Response

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Question

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A Sentiment Classification System is being developed to categorize text inputs into 5 sentiment categories (e.g., Very Positive, Positive, Neutral, Negative, Very Negative). The network is designed with the following specifications: **[5 Marks]**

- Input Features: 1,000
- Hidden Layer: Single hidden layer with only 8 neurons
- Output Categories: 5

Answer the following:

- a) Calculate the total number of trainable parameters in this network, excluding biases. Show your working clearly for each layer transition.
- b) After training, the model shows signs of overfitting — performing well on training data but poorly on unseen data. Critically analyze how the following architectural choices are contributing to this problem:
- The narrow width of the hidden layer (only 8 neurons)
 - The use of only a single hidden layer (shallow depth)
- c) Recommend the most suitable activation function for the hidden layer and the output layer of this architecture. Justify your choice for each, considering the nature of the task (multi-class sentiment classification).

Response

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Question

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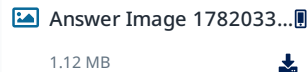
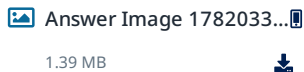
A Medical Diagnostic System classifies patient reports into 4 disease categories: Diabetes, Hypertension, Asthma, and Healthy. For a given patient's test report, the network produces the following logit scores: $Z = [1.6, 3.0, -0.5, -0.9]$ **[5 Marks]**

Class Order: [Diabetes, Hypertension, Asthma, Healthy]

- a) Compute the Softmax probability $\hat{y}$ for each disease category.
- b) The true diagnosis for this patient is "Hypertension" (2nd class). Compute the Cross-Entropy Loss.
- c) The Fully Connected Neural Network(FCNN) for this diagnostic system has the following architecture (biases excluded): Compute the total trainable weights
- Input Features: 30
- Hidden Layer 1: 20 neurons
- Hidden Layer 2: 12 neurons
- Output Layer: 4 neurons

Response

Words : 0



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